



Sheet of PW N°5

Exercise 1

Write a C program allowing to display even numbers belonging to the interval [5, 40].

Exercise 2

Write a C program allowing to

- Read an integer n (force the user to enter a positive integer),
- Calculate and display the sum of positive integers from 1 to n.

Exercise 3

To determine the greatest common divisor (GCD) of two numbers, we perform the subtraction of the larger number from the smaller one, then replace the larger number with the difference, and repeat this process until the difference becomes zero. The GCD is then the last non-zero result.

- Write a C program that calculates the GCD of two numbers.

Example : Determine the GCD of 12 and 28.

$$28-12=16$$

$$16-12=4$$

$$12-4=8$$

$$8-4=4$$

$$4-4=0$$

The last non-zero result is 4, so the GCD of 12 and 28 is 4.

Exercise 4

Write a C program allowing to display the following menu :

```
> Treatment 1
Treatment 2
Treatment 3
Treatment 4
```

The symbol '>' is initially positioned in front of 'Treatment 1'. The 'Up' and 'Down' direction keys allow to move this symbol to other treatments. If the 'Enter' key is pressed, the program displays the number of the selected treatment (Enter key code: 13).

Exercise 5

key = getch() ; is an instruction that allows to read a single character from the keyboard and assign it to the variable key The key codes required for our program are provided in the following table

Direction keys	
Top : 72	Right : 77
Down : 80	Left : 75
Esc : 27	

Complete the given C program in order to implement:

A star moves in 4 directions (right, down, left, up). Initially, the star is at coordinates (10, 10), and its direction is to the right,



If the movement is along the X-axis (to the right or to the left), the two directional keys, right and left, have no effect. In this case, if the user presses the up key (or down key), the direction of movement changes upward (or downward),

If the movement is along the Y-axis (upward or downward), the two directional keys, up and down, have no effect. In this case, if the user presses the right key (or left key), the direction of movement changes to the right (or to the left),

A point with random coordinates is marked by the symbol 'o'. If the star passes through the point 'o', the score increments by 1, and a new 'o' point is displayed,

The ESC (Escape) key is used to exit the application.

The following commands are required :

Function	Role
Sleep(T)	Stop execution for T milliseconds
system("cls")	Clear the screen
rand()	Generate random values
kbhit()	Indicate whether a key has been pressed
key=getch()	Read the character of the pressed key

The given C-program :

```
#include<stdio.h>
#include<windows.h>
#include<conio.h>
#include<stdlib.h>
```

```
void gotoxy(int x, int y)
```

```
{
    COORD c;
    c.X = x;
    c.Y = y;

    SetConsoleCursorPosition(
        GetStdHandle(STD_OUTPUT_HANDLE), c);
}
```

```
main(){
    X = 10;
    Y = 10;

    score = 0;
    Sens = 1;

    do {
```



```
switch (Sens) {  
    case 1 : {  
        X = X+1;  
        if (key == 80) Sens = 2;  
        else if (key == 72) Sens = 4;  
        break;  
    }  
    case 2: {  
    }  
    case 3:{  
    }  
    case 4:{  
    }  
}  
Sleep(100);  
} while (key != 27);  
}
```